Introduction

SWAT+ Documentation

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Contents

Pages . . . 1
What it computes . . . 1
What it is used for . . . 2
How it is run . . . 2
Source and license . . . 2
Code . . . 2
Inputs . . . 3
Routing . . . 3
Management . . . 3
Aquifers . . . 3
Outputs . . . 3
Migration . . . 4
License . . . 4
How to cite . . . 4
Source . . . 4
Versioning . . . 4
Tagging policy . . . 5
Where to find what changed . . . 5

SWAT+ is a watershed model. It simulates water, sediment, nutrient, pesticide, and crop yield response to climate and management at scales from a single field up to a large river basin.

This section covers what the model is, how it differs from the original SWAT, how to cite it, and how releases are numbered.

Pages

- What is SWAT+
- SWAT vs SWAT+
- Citing and license
- Release notes

The Soil and Water Assessment Tool Plus (SWAT+) is a public-domain watershed model. It is jointly developed by the USDA Agricultural Research Service (USDA-ARS) and Texas A&M AgriLife Research, with contributions from Colorado State University, the USDA-ARS Hydrology and Remote Sensing Laboratory, and other institutions.

SWAT+ simulates the quantity and quality of surface water and groundwater at scales from small watersheds to large river basins. It is used to evaluate the effects of land use, land management practices, soil, climate, and climate change on water resources.

What it computes

For each hydrologic response unit (HRU), routing reach, aquifer, reservoir, and wetland in a watershed, SWAT+ computes a daily mass balance for:

- water (precipitation, evapotranspiration, surface runoff, lateral flow, percolation, return flow, channel routing)

- sediment (detachment, transport, deposition)
- nutrients (nitrogen and phosphorus cycling in soil, plant uptake, transport in surface and subsurface flow, in-stream transformations)
- carbon (soil organic carbon pools)
- pesticides (application, decay, transport)
- salts and other constituents
- plant growth (biomass, yield, water and nutrient stress)

Output is reported daily, monthly, yearly, or as an annual average.

What it is used for

Typical applications:

- assessing the effect of management practices on water yield and water quality
- evaluating soil erosion and non-point source pollution
- estimating crop yield response to climate and management
- supporting watershed planning and total maximum daily load (TMDL) studies
- climate change impact studies on water resources

How it is run

SWAT+ is a Fortran command-line executable. It reads a folder of plain-text input files that describe the watershed, climate, soil, land use, management, and simulation period. It writes output as plain-text files in the same folder.

Most users do not write the input files by hand. They build a project with one of the supporting tools:

- **QSWAT+** is a QGIS plugin that delineates the watershed from a DEM, creates HRUs, and writes the input files.
- **SWAT+ Editor** is a desktop application for inspecting and editing input parameters in an existing project.

After the tools write the project folder, the SWAT+ executable is run from that folder.

Source and license

The source code is hosted at https://github.com/swat-model/swatplus and released under LGPL-2.1.

SWAT+ is the modernised successor to SWAT (Soil and Water Assessment Tool). The science is largely the same. The code structure, input file layout, and connectivity model are different.

The SWAT (2012) column below is sourced from the legacy model repository at `swat-model/swat2012`.

Code

	SWAT (2012)	SWAT+
Language	Fortran (fixed-form)	Fortran free-form (`.f90`)
Structure	Large shared common blocks	Modular, files grouped per process (`aqu_*`, `channel_*`, `hru_*`, `gwflow_*`, …)
Source	Distributed as archives	Git on GitHub, public, LGPL-2.1
Build	Hand-written Makefiles	CMake

The CMake build for SWAT+ uses `-free` (Intel) or `-ffree-line-length-none` (GNU) and glob-compiles `src/*.f90` into a single executable. See `CMakeLists.txt` in the source repository.

Inputs

SWAT2012 used a set of input file extensions tied to subbasins (`.sub`, `.rte`, `.gw`, `.sol`, etc.) with one file per HRU and subbasin.

SWAT+ replaces this with an object-based layout. The watershed is a graph of objects (HRUs, channels, aquifers, reservoirs, point sources, recall objects, decision tables) that are wired together through connectivity files (`.con`). The connectivity slots are defined by the `input_con` type in `src/input_file_module.f90`:

```
hru.con, hru-lte.con, rout_unit.con, gwflow.con, aquifer.con,
aquifer2d.con, channel.con, reservoir.con, recall.con, exco.con,
delratio.con, outlet.con, chandeg.con
```

Inputs are grouped by purpose, not by spatial unit:

- climate: `*.cli`, `*.pcp`, `*.tmp`, `weather-sta.cli`, `weather-wgn.cli`
- land use management: `landuse.lum`, `management.sch`, `cntable.lum`, `cons_practice.lum`, `ovn_table.lum`
- soils: `soils.sol`, `nutrients.sol`
- decision tables: `lum.dtl` (rule-based management triggers)
- hydrology: `hydrology.hyd`, `topography.hyd`
- basin-level: `codes.bsn`, `parameters.bsn`, `time.sim`, `print.prt`
- master index: `file.cio`

Routing

SWAT2012 routed water from subbasin to subbasin in a fixed order based on subbasin numbering.

SWAT+ uses the `*.con` files to describe arbitrary connectivity between HRUs, routing units, channels, aquifers, reservoirs, and exit points. Routing follows the graph defined in those files. This makes it possible to route landscape units (LSUs) within an HRU to a channel via a routing unit, or to send flow to a wetland or reservoir before the channel.

Management

SWAT2012 used `mgt` files with a fixed sequence of operations by date and heat unit.

SWAT+ separates the schedule (`management.sch`) from rule-based triggers (`*.dtl`, decision tables). A decision table evaluates conditions (soil water, plant heat units, day of year, etc.) at each time step and fires an operation when the condition is met. The same schedule can be reused across HRUs with different decision tables driving when operations occur.

Aquifers

SWAT2012 had a single shallow-aquifer storage per subbasin.

SWAT+ supports 1D aquifer objects, defined in `src/aquifer_module.f90` and read by `src/aqu_read.f90`, with one aquifer per landscape unit or group. It also includes a 2D groundwater module rooted in `src/aqu2d_read.f90`, `src/aqu2d_init.f90`, and the `gwflow_*` source files, where the aquifer is discretised in plan view and exchanges water laterally with channels, the unsaturated zone, reservoirs, and other objects.

Outputs

SWAT2012 wrote a fixed set of files (`output.hru`, `output.sub`, `output.rch`, `output.std`).

SWAT+ writes one file per object type and aggregation level, controlled by `print.prt`. Examples from `refdata/Ames_sub1`:

```
basin_wb_aa.txt        basin water balance, annual average
hru_wb_aa.txt          HRU water balance, annual average
```

```
basin_cha_aa.txt       channel, annual average
basin_aqu_aa.txt       aquifer, annual average
```

The aggregation suffix is `_day`, `_mon`, `_yr`, or `_aa`. Each object row in `print.prt` toggles `daily`, `monthly`, `yearly`, and `avann` independently with `y` or `n`.

Migration

There is no automatic converter from a SWAT2012 project to a SWAT+ project. The recommended path is to rebuild the project in QSWAT+ from the same DEM, soils, land use, and climate inputs, then transfer calibrated parameter changes by hand.

License

SWAT+ is released under the GNU Lesser General Public License, version 2.1 (LGPL-2.1). The full text is in the `LICENSE` file in the source repository.

How to cite

Cite the software release using the DOI:

> Arnold, J.G., Čerkasova, N., White, M.J., Bailey, R., Thorp, K., Jeong, J., Zhang, X., Ugraskan, T., van Griensven, A., Rathjens, H., Raj, C., Cai, X., Geter, W.F., David, O., Carlson, J.R., Le, K.N. *Soil and Water Assessment Tool Plus (SWAT+).* https://doi.org/10.5281/zenodo.14983534

The DOI `10.5281/zenodo.14983534` resolves to the current release; each tagged release also has its own DOI on Zenodo.

The machine-readable citation lives in `CITATION.cff` in the source repository. GitHub renders a "Cite this repository" button from this file.

Source

- Code: https://github.com/swat-model/swatplus
- Older versions (pre-GitHub): https://bitbucket.org/blacklandgrasslandmodels/modular_swatplus

SWAT+ releases are tagged on the GitHub releases page. Each tag corresponds to a Zenodo DOI under the concept DOI `10.5281/zenodo.14983534`.

Versioning

Releases are identified by a git tag. Tag names look like `61.0` or `61.1` for the current series, and earlier revisions in the reference data appear as `60.5.7`. The tag is read at configure time by `git describe --tags` (see `CMakeLists.txt`) and projected into the executable name and into the banner string in `main.f90`.

The executable name produced by CMake follows the pattern:

`swatplus-<tag>-<compiler>-<arch>[-Rel|-Dbg]`

where `<compiler>` is `gnu`, `ifo` (Intel `ifort`), or `ifx` (Intel LLVM), and `<arch>` is the host architecture. On Windows the executable has a `.exe` extension. This is defined in `CMakeLists.txt` as:

```
set(SWATPLUS_EXE "swatplus-${SWAT_VERSION}-${FFC}-${AR}${TY}")
```

A build between tags resolves to a `git describe` string of the form `<tag>-<n>-g<sha>`, for example `61.1-1-g92b4413`. See `doc/Tagging.md` in the source repository.

Tagging policy

See Tagging and releases in the developer guide for the rules a contributor follows when cutting a new tag.

Where to find what changed

For each release, the GitHub release page lists merged pull requests since the previous tag. The full commit history is available via:

```
git log <previous-tag>..<current-tag> --oneline
```

A change log curated for users (input file changes, output format changes, behaviour changes) is maintained in this section as releases are cut.